FinalProject

```r
# downloading the data files
library(xfun)
download_file(url = "https://raw.githubusercontent.com/egmccardy/ESP106FinalProject/refs/heads/main/fis
```

```
## [1] "fish-catch-gear-type.csv"
```

```r
download_file(url = "https://raw.githubusercontent.com/egmccardy/ESP106FinalProject/refs/heads/main/fis
```

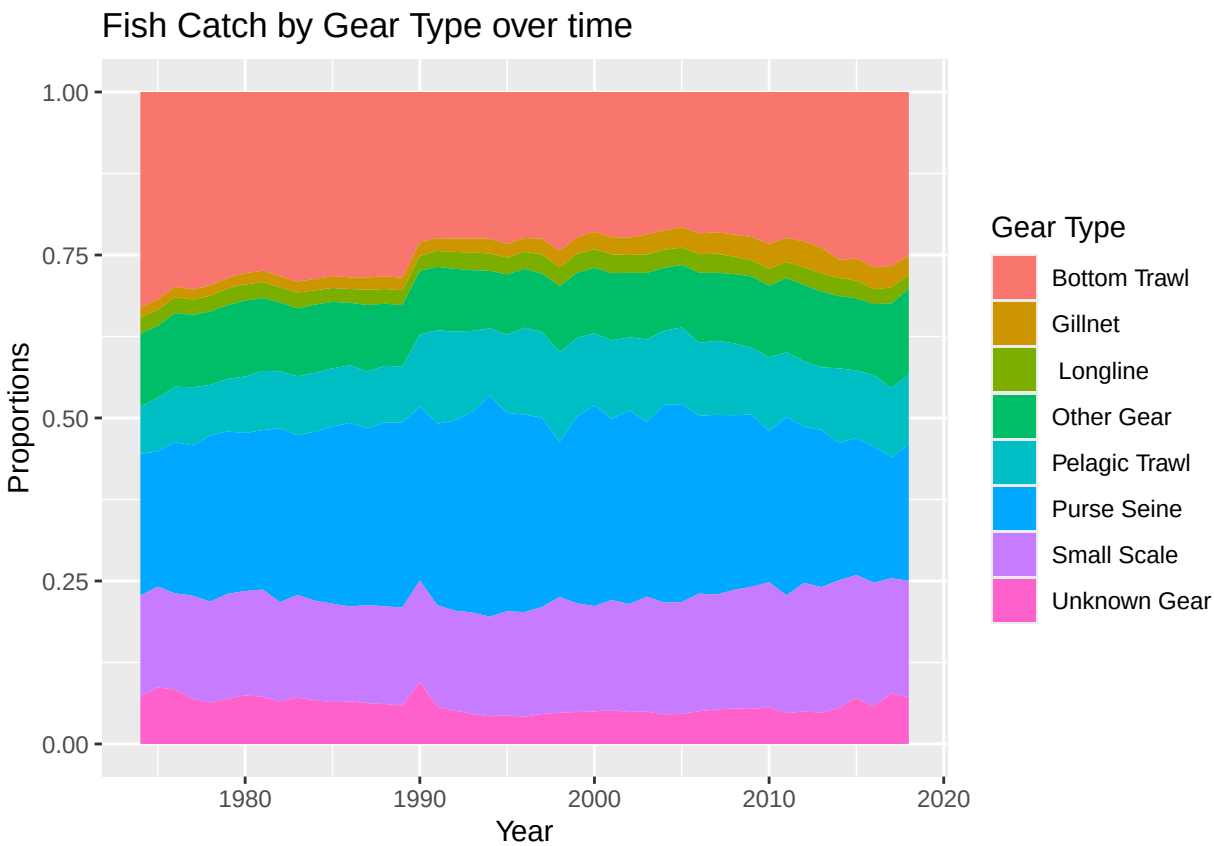
```
## [1] "fish-stocks-within-sustainable-levels.csv"
```

```r
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.0     v readr     2.1.6
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.2     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.1
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter()     masks stats::filter()
## x dplyr::lag()        masks stats::lag()
## x stringr::str_wrap() masks xfun::str_wrap()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become error
```

```r
library(ggplot2)
# Reading in the gear data
geartypes<-read.csv("fish-catch-gear-type.csv")
geartypesdf<-data.frame(geartypes)
wrldgear0<-filter(geartypesdf,Entity=="World")
wide_data<- data.frame(Year=wrldgear0$Year,Unknown.Gear=wrldgear0$Unknown.gear,other.gear=wrldgear0$Oth
# Converting the data frame to long format in order to plot
wrldgear.long<-wide_data %>%
  pivot_longer(cols=c(Unknown.Gear, other.gear,longline,gillnet,small.scale,purse.seine,pelagic.trawl,b
# Code for the gear use over time plot
 ggplot(wrldgear.long,aes(x=Year,y=Count,fill=Gear.Type)) + geom_area(position= "fill")+labs(title="Fis
```

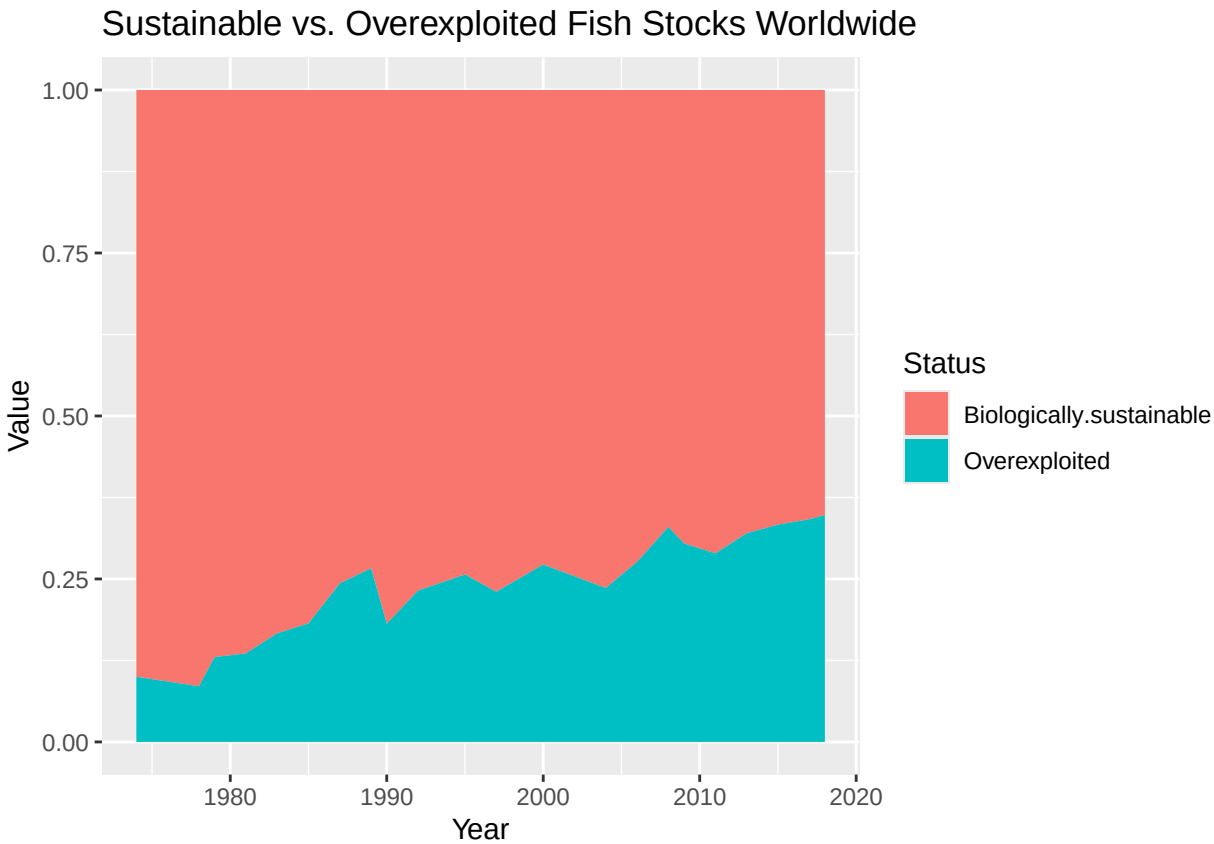
```
## Warning: Removed 192 rows containing non-finite outside the scale range
## (`stat_align()`).
```



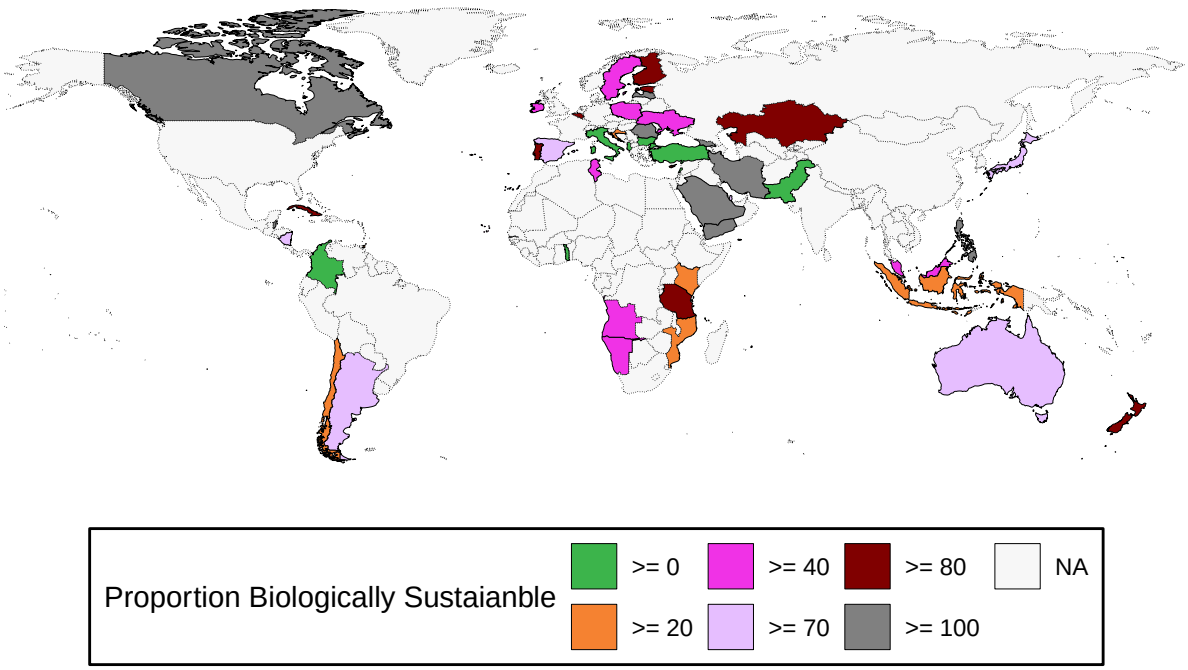

```
# Reading in the data on fish stock classifications
fish.stock.sustainable<- read.csv("fish-stocks-within-sustainable-levels.csv")
wrldstocks0<- data.frame(fish.stock.sustainable)
wrldstocks<- filter(wrldstocks0, Entity == "World")
colnames(wrldstocks)<- c( "Entity", "Code", "Year", "Proportion1", "Proportion")
wrldstocks$Proportion<- wrldstocks$Proportion1/100
wrldstocks<- wrldstocks[,-c(1,2,4)]
# Converting the counts into proportions
fractions<-wide_data[25:69,]%>%
  rowwise()%>%
  mutate(across(Unknown.Gear:bottom.trawl, ~./sum(c_across(Unknown.Gear:bottom.trawl))))%>%
  ungroup()
wrldstocks2<- wrldstocks0[1:45,]
# Establishing a data frame with gear and fish stock classifications proportions worldwide
fractions$Biologically.Sustainable<- wrldstocks2$Proportion
colnames(fractions)<- c("Year", "Unknown Gear", "Other Gear", "Longline", "Gillnet","Small Scale", "Pur
```

```
library(tidyverse)
library(ggplot2)
# plotting a stacked 100% area chart of the proportion of fish stocks classified as biologically sustai
stock_sustdf <- data.frame(fish.stock.sustainable)
wrld_sust <- filter(stock_sustdf, Entity == "World")%>%
  filter(Year >= 1974 & Year <= 2018)

wrld_sust_long <- wrld_sust %>%
  pivot_longer(
    cols = c(Overexploited, Biologically.sustainable),
```

```r
    names_to = "Status",
    values_to = "Value"
  )

ggplot(wrld_sust_long,
       aes(x = Year, y = Value, fill = Status)) +
  geom_area(position = "fill") +
  labs(
    title = "Sustainable vs. Overexploited Fish Stocks Worldwide",
    fill = "Status"
  ) +
  scale_x_continuous(limits = c(1974, 2018))
```

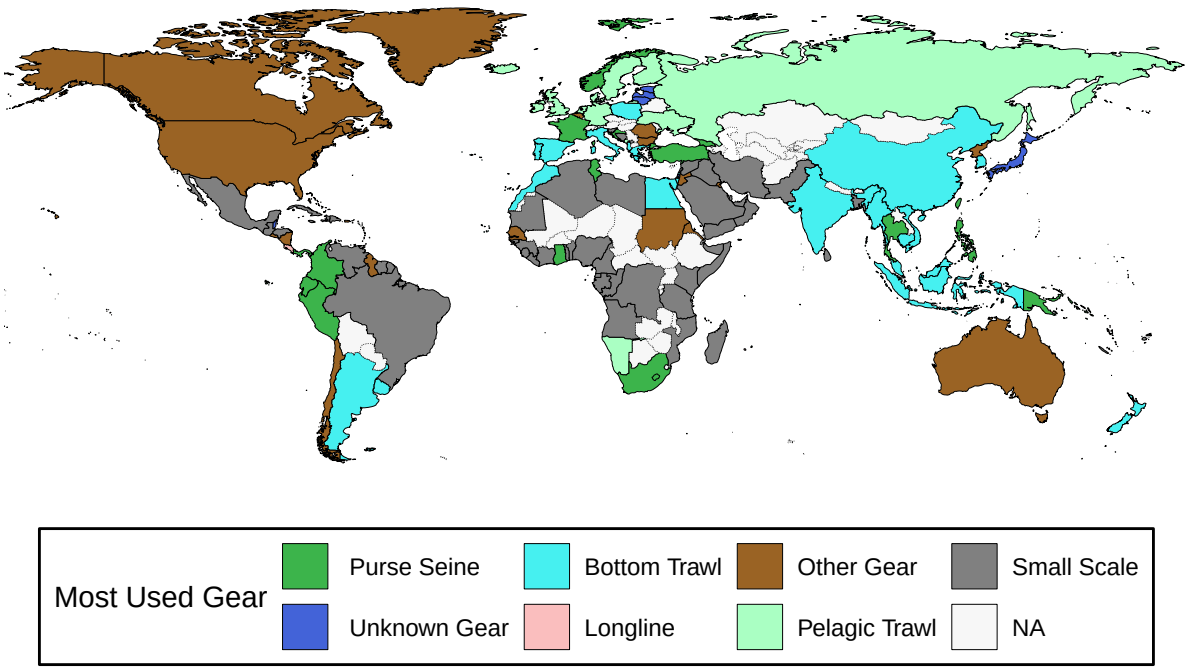



```r
# plotting the proportion of fish stocks classified as biologically sustainable by country in 2018
library(countries)
 fish.stock.sustainable.2018<- filter(data.frame(fish.stock.sustainable), Year == "2018")
 fish.stock.sustainable.2018<- fish.stock.sustainable.2018[-55,-c(3,5)]
 colnames(fish.stock.sustainable.2018)<- c("Entity", "Code", "Proportion Biologically Sustaianble")
quick_map(data= fish.stock.sustainable.2018, plot_col = "Proportion Biologically Sustaianble", theme= "
```




```
# plotting most used gear by country in 2018
gear.2018<- filter(geartypesdf, Year == "2018")
gear.2018<-gear.2018[,-3]
colnames(gear.2018)<- c("Entity", "Code", " Unknown Gear", "Other Gear", "Longline", "Gillnet", "Small S
gear.max<- apply(gear.2018, 1, function(x) colnames(gear.2018)[which.max(x)])
gear.max.df<- data.frame(gear.max)
gear.max.df$country<- gear.2018$Entity
gear.max.df$Code<- gear.2018$Code
colnames(gear.max.df)<- c("Most Used Gear", "Country", "Code")
quick_map(data= gear.max.df, plot_col= "Most Used Gear", theme= "Distinct")
```

```
## Some country IDs have no match in one or more of the requested country naming conventions, NA returne

## Multiple country IDs have been matched to the same country name.

## There is low confidence on the matching of some country names, returning the closest match.

##
## Set - verbose - to TRUE for more details
```

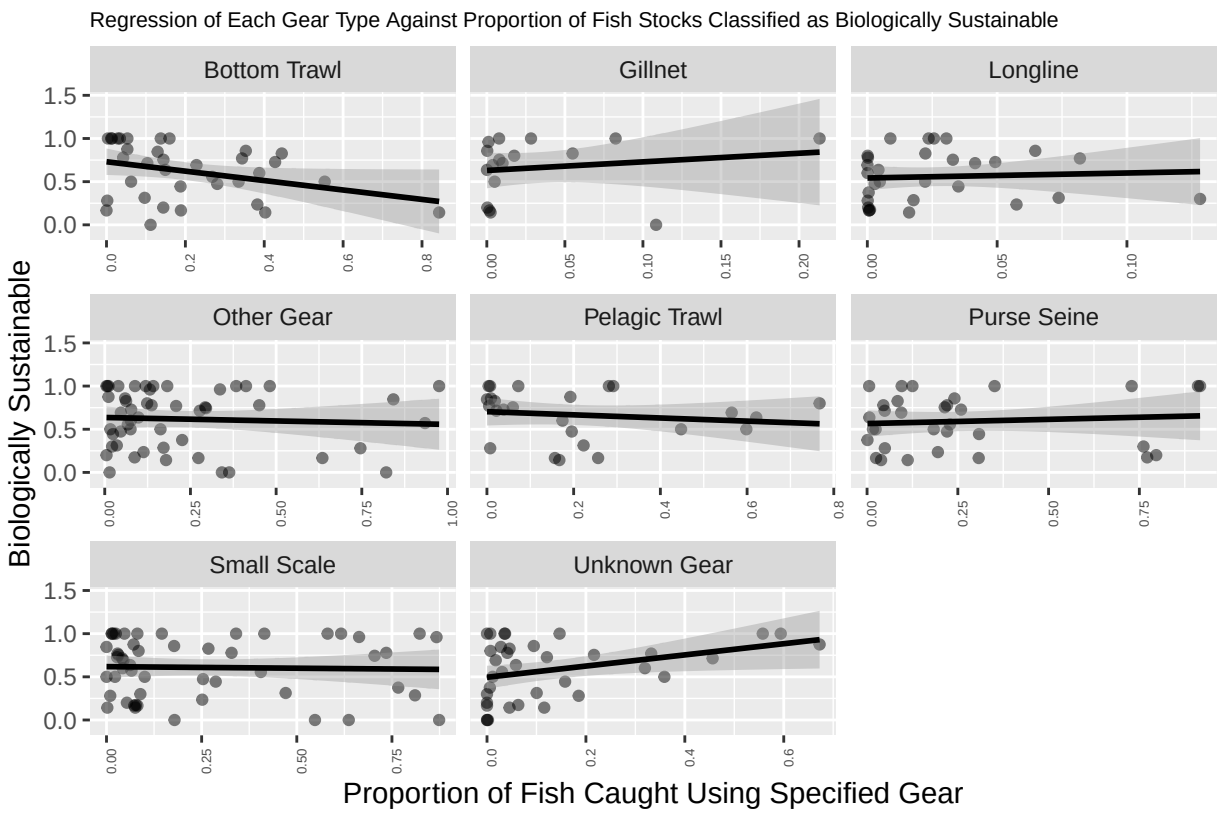



```
# creating a data frame showing proportions by country of both gear used as classification of fish stoc
wrldgear.countries<- filter(geartypesdf, Entity != "World")
wrldgear.countries<- filter(wrldgear.countries, Year == "2018")
wide_data2<- data.frame(Entity=wrldgear.countries$Entity, Unknown.Gear=wrldgear.countries$Unknown.gear,
# Converting counts to fractions
fractions2<-wide_data2%>%
  rowwise()%>%
  mutate(across(Unknown.Gear:bottom.trawl, ~./sum(c_across(Unknown.Gear:bottom.trawl), na.rm= TRUE)))%>%
  ungroup()
# Merging the data on gear used and classifications of fish stocks into one data frame
fractions3<- inner_join(fractions2, fish.stock.sustainable.2018, by= "Entity")
 fractions3<- fractions3[,-10]
 fractions3$`Proportion Biologically Sustainable`<- fractions3$`Proportion Biologically Sustaianble`/10
colnames(fractions3)<- c("Entity","Unknown Gear", "Other Gear", "Longline", "Gillnet", "Small Scale", "
# Running Pearson's Chi- Squared Test on gear type used in fishing against proportion of fish stocks cl
gear.cols2<- c("Unknown Gear", "Other Gear", "Longline", "Gillnet", "Small Scale", "Purse Seine", "Pela
results2<- do.call(rbind, lapply(gear.cols2, function(gear){
  test<- cor.test(fractions3[[gear]], fractions3$`Proportion Biologically Sustainable`)
   data.frame(
    Gear    = gear,
    r       = round(test$estimate, 3),
    t       = round(test$statistic, 3),
    df      = test$parameter,
    p.value = round(test$p.value,4)
   )
}))
rownames(results2)<- c("Unknown Gear", "Other Gear", "Longline", "Gillnet", "Small Scale", "Purse Seine
results2<- results2[,-c(1,6)]
results2$p.value<- format(results2$p.value, scientific = TRUE, digits= 3)
```

```
# Constructing a linear regression of gear type against proportion of fish stocks classified as biologi
fractions3<- fractions3[,-c(1,10)]
colnames(fractions3)<- c("Unknown Gear", "Other Gear", "Longline", "Gillnet","Small Scale", "Purse Sein
```

```
response_var<-names(fractions3)[ncol(fractions3)]
fractions.for.plot<- fractions3 %>%
  pivot_longer( cols= -c(all_of(response_var)), names_to = "Predictor", values_to= "Value") %>%
  ggplot(aes(x= Value, y= .data[[response_var]])) + geom_point(alpha= 0.5)+geom_smooth(method="lm", se=
print(fractions.for.plot)
```

```
## `geom_smooth()` using formula = 'y ~ x'

## Warning: Removed 138 rows containing non-finite outside the scale range
## (`stat_smooth()`).

## Warning: Removed 138 rows containing missing values or values outside the scale range
## (`geom_point()`).
```




```
# plotting the proportion of fish stocks classified as biologically sustainable by country in 2021
fish.stock.sustainable.2021<- filter(data.frame(fish.stock.sustainable), Year == "2021")
 fish.stock.sustainable.2021<- fish.stock.sustainable.2021[-55,-c(3,5)]
 colnames(fish.stock.sustainable.2021)<- c("Entity", "Code", "Proportion Biologically Sustaianble")
quick_map(data= fish.stock.sustainable.2021, plot_col = "Proportion Biologically Sustaianble", theme= "
```

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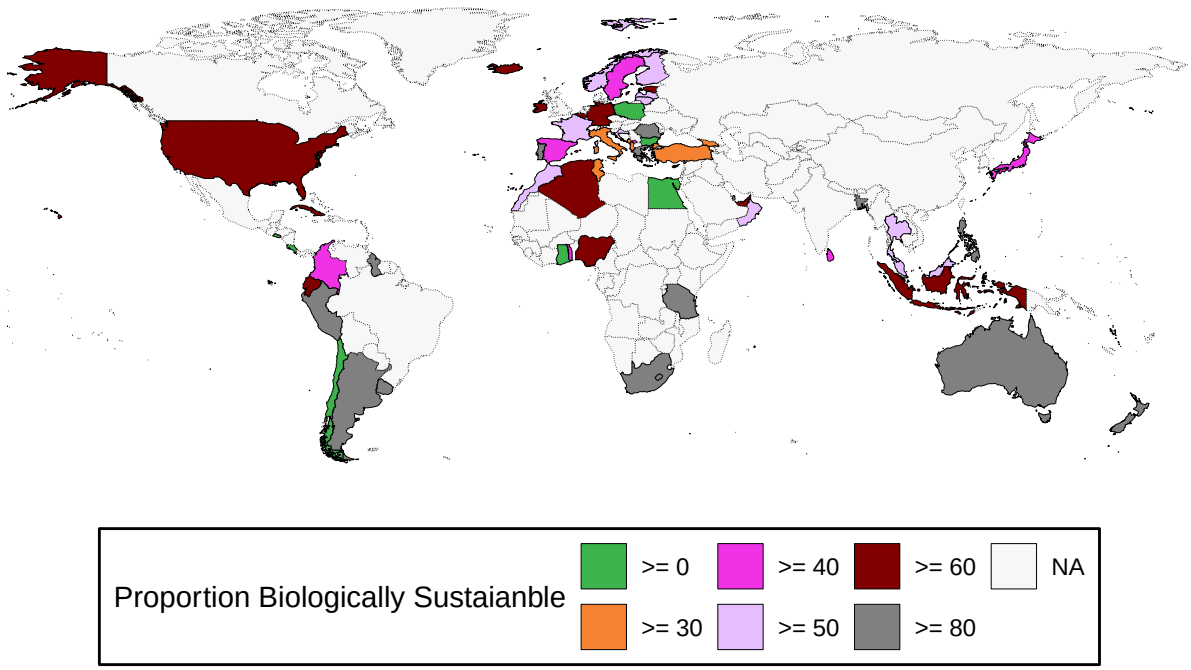


```
#Identifying the countries with the highest total fish catch in 2018
fish.caught.df<- filter(geartypesdf, Year == "2018")
fish.caught.df<- fish.caught.df[, -3]
fish.caught.df$total.catch<- rowSums(fish.caught.df[,c("Unknown.gear", "Other.gear", "Longline", "Gilln
fish.caught.total<- fish.caught.df[,-c(3:10)]
colnames(fish.caught.total)<- c("Entity", "Code", "Total Catch")
fish.caught.total<- fish.caught.total[-197,]
fish.caught.total<- fish.caught.total %>%
  arrange(desc(`Total Catch`))
head(fish.caught.total)
```

```
##          Entity Code Total Catch
## 1         China  CHN    14336047
## 2        Russia  RUS     8252283
## 3          Peru  PER     7830707
## 4     Indonesia  IDN     6753766
## 5 United States  USA     6427671
## 6      Thailand  THA     4730454
```